Minecraft Economics: A Study of Wealth Inequality in a Virtual World

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Abstract

We examine the hypothesis that capitalist economies generate inequality by looking at the distribution of wealth in a virtual world. Employing the sandbox video game Minecraft as an example of a "pure" market economy, we chose several Minecraft economy servers and collected their respective wealth data. We found these unregulated economies to have greater wealth inequality than any single nation's economy. These results suggest that market economies generate significant "natural" inequality, and that government policies are effective in moderating the inequality-inducing impact of capitalism.

Keywords: Inequality, Wealth, Gini coefficient, virtual economy, Minecraft

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1. Introduction

A recurring question in economics concerns how much inequality capitalist economies create "naturally" (i.e., without government intervention). Some authors, (e.g. (Kuznets 1955)) have argued that capitalist economies tend to reduce inequality, while others ((“Ricardo, On the Principles of Political Economy and Taxation | Library of Economics and Liberty” n.d.); (Marx 1887); and (Piketty 2014)) have argued the opposite. The issue is a difficult one to address empirically for at least two reasons. First, the unit of analysis (typically a country) is subject to many factors that can affect inequality. For example, numerous parts of the tax code in modern economies exist to reduce inequality; however, given the complexity of modern tax codes, it is challenging to determine how individual requirements impact inequality. Relatively sparse data means there is a high ratio of explanatory factors to data points, making empirical analysis challenging. Second, although there is relatively good data on income, in many countries it is difficult to gather data on wealth. For example, there is no complete data set on the wealth distribution in the United States; instead, researchers must rely on incomplete data sets that sometimes generate conflicting conclusions (Kopczuk 2015).

This paper attempts to address how inequality evolves in market economies by addressing the two common challenges listed above. We examine the distribution of wealth in a market environment with minimal government intrusion, and in an environment in which we are able to know exactly how much wealth each individual holds. Although these economies come from video games, the richness of the economic interactions that are available to players, combined with the low regulatory environment and extremely high quality data, allow us to begin to draw some conclusions regarding the evolution of wealth distribution. We find that in these "pure" market economies, wealth inequality is higher than inequality in low inequality economies (like Japan), than in high inequality economies (like the United States), and even higher than a global measure of inequality. We interpret this finding as support for the

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claim that markets do tend to generate inequality, but also as evidence that governments' policies to reduce inequality are effective.

2. Methodology

To assess the "natural" evolution of inequality under "pure" capitalism, one needs to find an economy with no redistributive policies. Such economies do not exist in the real world, but sometimes exist in virtual worlds. The virtual world that will be used in this study is Minecraft - a sandbox video game created in 2011 by Swedish programmer Markus "Notch" Persson and published by Mojang (now available at minecraft.net).

Readers may be skeptical that data from a virtual world is useful for understanding "real" economies. However, (Castronova et al. 2009; Castronova, Knowles, and Ross 2015) argue that virtual economies tend to behave like other, "real" economies. In fact, researchers have looked at virtual economies to examine a number of economic phenomenon, including GDP identities and the quantity theory of money (Castronova et al. 2009) and the evolution of unregulated capital markets (Bloomfield and Cho 2011).

The Minecraft video game allows players to meet virtually on "servers," where a wide variety of activities take place. Players can chat or play games with one another. The particular type of server that we are interested is labeled "economy" by Mojang, and has the following basic set of rules. The primary goal of the players on the server is to advance in rank, which can be purchased with gold, which functions as these economies' unit of account and exchange. Players interact primarily in two types of environments: mines and markets. In mines, players are able to employ tools to "mine" resources like diamonds or iron ore. These resources have a degree of fungibility; some can be turned into tools or armor, and all can be bought and sold in markets. In some types of mines, players merely collect resources. In other mines, labeled PVP ("Player vs Player"), players are able to interact with each other

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violently. For example, in a PVP mine player A could attack player B and take all the resources on B's person. In markets, players are able to buy and sell resources, and are able to convert gold into rank. Unlike in mines, in markets the rules of the game do not allow for violence or involuntary exchange; that is, property rights are perfectly enforced. (Here we see the application of Lessig's observation that in virtual economies, "code is law" (Lessig 2008).) Although each server may have custom rules, the basic observation is that each server has an economy, with a standardized unit of exchange, and with well-defined rules for making market transactions. Importantly, there are no redistributive rules - each player is responsible for his/her own resources, and transfers of resources from one player to another are never made by individuals running the server.

2.1. Data Collection

Data was collected through the game Minecraft, using the in-game command "/baltop", which shows the current gold balance of every player in that server. The players were ranked based on their numerical wealth value, i.e. larger values first, which were displayed in on pages, with nine wealth values per page. The quantity of pages depends on the size of the server. The nine servers were chosen randomly from http://minecraft-mp.com/; however, the "economy tag" had to be the foremost tag detailing the server. The data (on 17,676 players) was collected between July 12, 2016 and September 15, 2016.

2.2. Gini Coefficient

We use the Gini coefficient to measure wealth inequality. Its value ranges from 0 to 1, 0 the value of perfect equality and 1 the value of maximum inequality. Numerous formulas for the Gini coefficient abound, but given the nature of the data we wish to analyze, we will use the following formula (from (Deaton 1997)):

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$$Gini = \frac{N+1}{N-1} - \frac{2}{N(N-1)\mu}\left(\sum_{i=1}^{N} P_i Y_i\right)$$

where N is the number of observations, μ is the mean wealth, P_i is the income rank P of person i, with an income of Y, in a way that the poorest individual receives a rank of N and the richest a rank of 1.

3. Results

Table 1 gives the descriptive statistics for all of the servers. The number of players across servers varies considerably, as does average wealth. This second observation comes from the fact that the "price level" of the server is set idiosyncratically by the server's owner - some servers have high price levels, (reflecting a large money supply), while other servers have lower price levels (again, reflecting a lower money supply). As is the case in the real world, the distribution of wealth is highly positively skewed, with the standard deviation of wealth typically being an order of magnitude greater than the mean. We see that the levels of inequality are high, with only one server having a Gini coefficient less than 0.9.

Table 1. Descriptive Statistics

Server	**Charlie**	**Evisea**	**Ezure**	**Gcraft**	**Norse**	**Odin**	**Pcraft**	**Sky Srvrs**	**Tempus**
# players	387	1333	1719	349	1223	3422	759	2770	5714
AVG wealth	2.5E1	5.1E3	1.5E6	8.1E4	6.5E4	8.8E6	2.0E9	1.6E4	9.0E6
STD DEV wealth	5.5E1	4.1E4	8.8E6	3.1E5	8.3E5	3.2E8	3.5E10	3.3E5	6.7E8
Max wealth	4.5E2	8.5E5	3.8E7	3.0E6	2.0E7	1.3E10	6.2E11	1.4E7	5.0E10
Gini	0.9262	0.9172	0.9794	0.8791	0.986	0.9995	0.9987	0.9503	0.9998

In table 2, we provide Gini coefficients from (Davies et al. 2009) on comparable economies, including their estimate of world-wide inequality. We also present data on inequality generated in the agent-based simulation Sugarscape (Epstein and Axtell 1996). Note that the Minecraft economies are more

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unequal that even the least equally distributed national economy (Namibia), and are more unequal that the world economy. The Minecraft economies are substantially more unequal than developed economies with low inequality (e.g., Japan) and high inequality (e.g., the United States), which have significant social safety nets and policies to mitigate inequality.

Table 2. Sample Gini Coefficients.

Unit	Gini
Minecraft Average	0.96
World	0.89
Namibia	0.85
United States	0.8
Sugarscape (inheritance)	0.75
Japan	0.54
Sugarscape (no inheritance)	0.5

4. Discussion

We have presented evidence that virtual economies lacking redistributive policies appear to generate more inequality than real economies. However, this finding must be tempered due to a number of potentially confounding factors. First, one must ask how well does a Minecraft economy mimic a real economy? Although the Minecraft marketplace allows for rich interactions between players, it is incredibly simple when compared to the real world. For example, there are no capital or insurance markets, and property rights are cartoonish, either fully enforced or not at all. In addition, the level of violence between players is much higher than in most real economies. Also, the ability of players to form coalitions or other sorts of contracts is absent. These factors may contribute to the high levels of inequality observed.

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Second, one must consider that the level of wealth inequality may have been (advertently or inadvertently) designed into the game. Games in which the outcome is the same for all players may not be considered entertaining, leading game designers to make decisions that increase inequality.

We can gain some more insight by considering how the Minecraft economies compare to the Sugarscape economy (Epstein and Axtell 1996). In their simple agent-based simulation, they generated only moderate amounts of inequality; however, they also showed how sensitive this inequality was to time. That is, in the Sugarscape economy initially all agents had similar levels of wealth, and as the agents were allowed to harvest and trade, more inequality arose. It seems likely that a similar process arises on the Minecraft servers. If we use the number of players on the server as a proxy for the length of time the server has been in existence, we see that larger numbers of participants are positively correlated (r=0.55) with the Gini coefficient.

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